

$\mathbb{R}^{2k}$ is Theoretically Large Enough for Embedding-based Top- k Retrieval

Minimal Embeddable Dimension

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Geometric approximability is not the obstruction.

The approximability question

Embedding retrieval stores objects as vectors, embeds a query, and retrieves by score comparison.

Minimal Embeddable Dimension

Given m objects, MED is the least d such that some vectors in $\mathbb{R}^d$ can exactly retrieve every answer set

$$S \subseteq X, \quad 1 \leq |S| \leq k$$

by score comparison and a threshold.

This isolates exact geometric capacity from positive-margin or learned embedding protocols.

Exact capacity is $\Theta(k)$, not a function of m

Scoring rule	Lower bound	Upper bound
Inner product	$k - 1$	$2k$
Euclidean distance	$k - 1$	$2k$
Cosine similarity	$k - 1$	$2k + 1$

- Lower bounds come from VC dimension.
- Upper bounds are explicit constructions.
- Exact geometric capacity does not grow with the universe size m .

The construction in one line

Place objects on the moment curve:

$$v_i = (t_i, t_i^2, \dots, t_i^{2k}).$$

For a target subset S , form

$$P_S(t) = \prod_{i \in S} (t - t_i).$$

Use the coefficients of $-P_S(t)^2$ as the query vector.

Closed-form witness

Selected objects score at the shared maximum; every other object scores lower.

Robust MED: normalized gaps change the regime

Exact MED is the no-margin problem.

RMED adds unit-norm vectors and a fixed selected-unselected score gap ϵ .

Positive gap makes m reappear through packing:

$$\text{RMED}(m, k, \epsilon) \geq \frac{\log \binom{m}{k}}{\log(1 + 2/\epsilon)}.$$

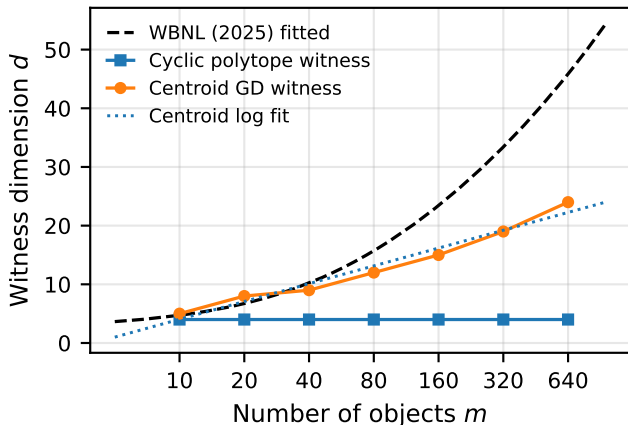
Finite- m gaps are capped by

$$\epsilon_{\star}(m, k) = \frac{m}{\sqrt{k(m-1)(m-k)}}.$$

The paper's RMED shorthand is the large-universe regime $m/k \rightarrow \infty$, where $\epsilon_{\star}(m, k) \sim 1/\sqrt{k}$. At the feasible $c/\sqrt{k}$ scale, a Gaussian centroid witness gives

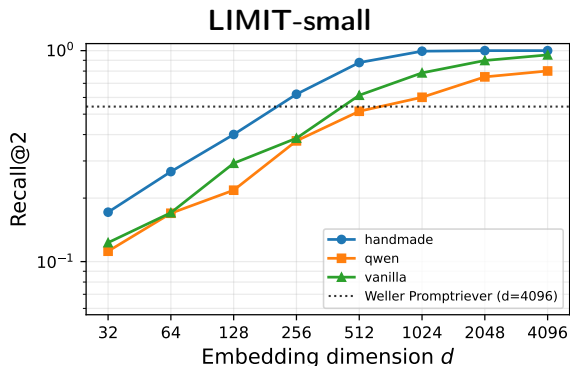
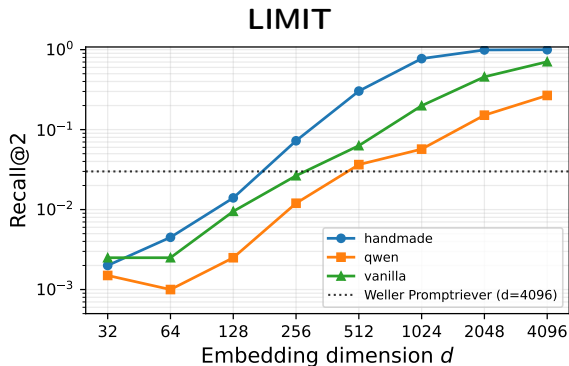
$$\text{RMED-C}(m, k, c/\sqrt{k}) \leq O(k^2 \log m).$$

Synthetic top-2: checked witnesses, not lower bounds



- Cyclic polytope gives an exact $d = 4$ witness for every tested m .
- Centroid GD and its log fit stay far below the WBNL fitted curve.
- A failed optimization run is not a proof of infeasibility.

LIMIT: simple single-vector baselines are strong



All three tokenizations cross the reported Promptriever line; vanilla is label-unaware and unsupervised. The packaged LIMIT top-2 tasks can also be exactly overfit in $\mathbb{R}^4$ by the cyclic-polytope construction.

What this means for retrieval

Exact MED

Cyclic-polytopal neighborliness gives $\Theta(k)$ dimensions for inner product, cosine, and Euclidean scoring, independent of m up to constants.

Robust MED

The normalized-gap problem is different: finite- m gaps are capped by $\epsilon_{\star}(m, k)$, and Gaussian centroid witnesses give $O(k^2 \log m)$ dimensions at the feasible large-universe scale.

Empirical interpretation

The synthetic and LIMIT failures are not failures of geometric approximability; the remaining difficulties lie in learning, tokenization, objectives, conditioning, finite precision, and optimization.